Blessed G/Michael Catholic Secondary School

First Semester Physics Test One for Grade 11

Time Allowed 1:20 hr

Dec, 2018 E.C

I. *Write "True" if the satmnet is correct or write "False" if the statement is wrong.* *[1 point each]*

True 1. Uniform motion and uniformly accelerated motion are straight line motions.

True 2. In Uniform Circular Motion , the instantaneous velocity and the centripetal acceleration are always perpendicular to each other $\lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t}$ or

False 3. An object is launched vertically upward from the ground. When it returns to the ground, its total displacement is zero, but its average velocity for the entire flight is non-zero.

True 4. Every time the position of an object undergoing one-dimensional motion reverses direction (a "turning point"), the corresponding velocity-time graph must cross the horizontal time axis (v=0) at that exact moment.

False 5. A ball thrown vertically upwards is in uniformly accelerated motion. At the peak of its flight, its instantaneous velocity is zero, and therefore, its instantaneous acceleration also be zero.

II. *Choose the best answer from the given alternative.* *[1 point each]*

D 6. Which of the following statements is a necessary condition for an object to be in uniform motion?

A. The object's speed must be constant, but its direction can change.

B. The object's total displacement must be zero.

C. The object must be at rest or moving in a straight line.

D. The object's acceleration must be zero and moving in a straight line.

B 7. A particle moves in a straight line for 20 seconds with velocity 3 m/s and then velocity 4 m/s for another 20 seconds and finally moves with velocity 5 m/s for next 20 seconds. What is the average velocity of the particle? 60+80+100=240 20m1=60 4m/s

A. 3 m/s B. 4 m/s C. 5 m/s D. Zero

A 8. A runner sprints 4.0 m East in 2.0 s, and then immediately turns around and jogs 3.0 m West in 5.0s. What is the magnitude of the runner's average velocity for the entire 7.0 second interval?

A. 0.14 m/s B. 0.7 m/s C. 1.0 m/s D. 1.3 m/s

B 9. Which of the following conditions is a necessary and sufficient requirement for an object to be considered in Uniformly Accelerated Motion?

A. speed must be constantly increasing. decrease

B. The object's velocity must change by the same amount in every equal time interval.

C. The object's instantaneous velocity and acceleration vectors must be in the same direction.

D. The object's velocity-time graph must be a parabola.

10. A car starts from rest ($v = 0$) and accelerates uniformly. If the car covers a displacement of x in time t, how much displacement will it cover in the next t time interval?

A. x B. 4x C. 3x D. 2x

A 11. The Ethiopian Airlines Boeing airplane lands on one of the Bole international airport runways with an initial velocity of 40.0 m/s and then decelerates at -2.0 m/s^2. What is the time taken to became at rest?

A. 20 s B. 40 s C. 60 s D. 80 s

B 12. A ball is thrown vertically upwards with a velocity of 10m/s from the balcony of a tall building. The balcony is 15m above the ground and gravitational acceleration is 10m/s^2. What is the time required for the ball to hit the ground?

A. 2 sec B. 3 sec C. 4 sec D. 5 sec

C 13. Suppose a planner is designing an airport for small airplanes. Such planes must reach a speed of 40 m/s before takeoff and can accelerate at 10.0 m/s^2. What is the minimum length for the runway of this airport?

A. 20 m B. 60 m C. 80 m D. 100 m

D 14. Which of the following statements about the velocity-time (v-t) graph of an object in uniform motion is correct?

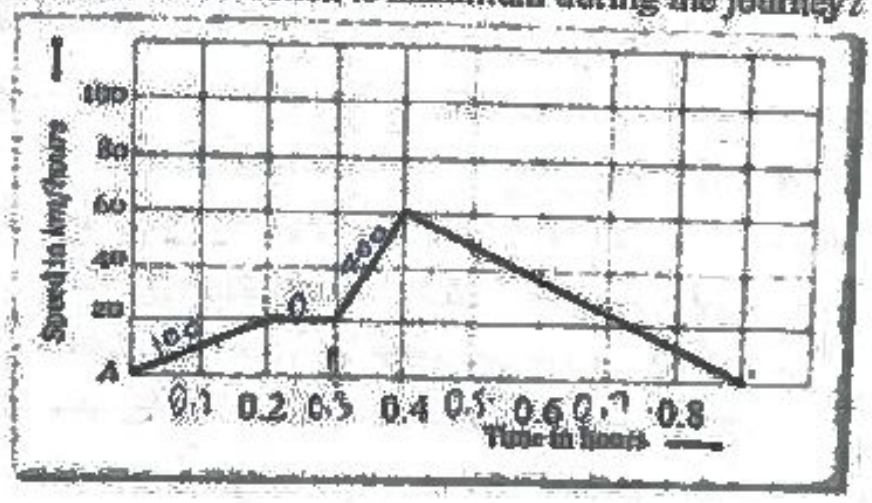
A. The graph cannot be a straight line if the object is actually moving.
B. The graph is a straight line with a constant, non-zero slope.
C. The area under the graph represents the object's acceleration.
D. The graph is a horizontal line with a slope of zero.

B 15. A particle is in motion and is accelerating. The velocity of the motion as a function of time is given by: $v(t) = (20t - 5t^2)$ m/s. Find the instantaneous acceleration at $t = 3$ s.

A. 10 m/s B. -10 m/s C. 15 m/s D. -15 m/s

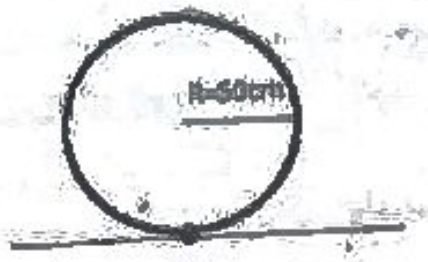
C 16. A train move from one station to another in 0.9 hours time. Its speed time graph during this motion is shown in the figure. At what time interval the acceleration is maximum during the journey?

A. 0 to 0.2 hr
B. 0.2 to 0.3 hr
C. 0.3 to 0.4 hr
D. 0.4 to 0.9 hr



B 17. Find the displacement of the point of the wheel initially in contact with the ground, when the wheel roles forward one full rotation?

A. 0.5π m
B. π m
C. 1.5π m
D. 2.0π m



A 18. A stone tied to a string is swung in a horizontal circle at a constant speed. At any given instant, the centripetal acceleration vector a_c of the stone is directed:

A. Inward, toward the center of the circle.
B. Tangentially, in the direction of the stone's motion.
C. In the direction opposite to the stone's instantaneous velocity.
D. Outward, away from the center of the circle.

B 19. An object moves in a circle of radius R=2.0 m and completes 5 full rotation in 10.0 s. What is the constant speed (v) of the object?

A. 4π m/s B. 2π m/s C. 10π m/s D. 20π m/s

A 20. A particle undergoing uniform circular motion has a period T and a constant speed v. If the particle's speed is doubled (v→2v), what is the new period, assuming the radius R remains constant?

A. $\frac{1}{2}T$ B. $\frac{1}{4}T$ C. $2T$ D. $4T$

III. *Work out (show all necessary steps) . 15 point totalf*

21. On its way along a straight line, a car starts from rests and accelerated at a rate of 2.5 m/s^2 for 10 seconds. It maintained its speed for further 5 seconds and decelerated by 0.5 m/s^2 come to rest.

a) What is maximum speed of car? 25 m/s
b) What is the time taken for the entire motion? 65 sec
c) What is the total displacement? 875 m

22. At the instant the traffic light turns green, an automobile starts moving with a constant acceleration of 5.0 m/s^2. At the same instant a truck, travelling with a constant speed of 10 m/s, overtakes and passes the automobile.

a) How far beyond the traffic signal will the automobile overtake the truck? 400 m
b) How fast will the automobile be at that instant? 20 m/s

25/25 Excellent

ANSWER SHEET

Name: Muhamed Menale Grade: 11 Section: C

Thank you R.No 03

1. True
2. True
3. False
4. True
5. False
6. D
7. B
8. A
9. B
10. C
11. A
12. B
13. C
14. D
15. B
16. C
17. B
18. A
19. B
20. A

21. Given

1st: $u_1 = 0$, $a_1 = 2.5\ m/s^2$, $t_1 = 10\ sec$

2nd: $t_2 = 5\ sec$

3rd: $a_3 = -0.5\ m/s^2$, $v_3 = 0$

a. $V_{max} = ?$ $V_{max} = u + at$
$= 0 + 2.5(10)$
$= 25\ m/s$

b. let $V_{max} = V_2$
$V_2 = V_1$
also $V_2 = u_3 = 0$
$u_3 = 25\ m/s$
$t_3 = \frac{V_3 - u_3}{a_3} = \frac{0 - 25}{-0.5} = \frac{250}{5} = 50\ sec$
$t_T = t_1 + t_2 + t_3$
$= (10 + 5 + 50)\ sec$
$= 65\ sec$

c. $S_1 = \frac{1}{2} a_1 t_1^2$
$= \frac{1}{2}(2.5)(10)^2$
$= 2.5 \times 100$
$= 1.25 \times 100$
$= 125\ m$

$S_2 = v_2 t_2$
$= 25\ m/s \times 5\ sec$
$= 125\ m$

$S_3 = u_3 t_3 + \frac{1}{2} a_3 t_3^2$
$= 25(50) + \frac{1}{2}(-0.5)(50)^2$
$= 1250 - 625$
$= 625\ m$

$S_T = S_1 + S_2 + S_3$
$= (125 + 125 + 625)\ m$
$= 875\ m$

22. Given

A: $u = 0$, $a = 5\ m/s^2$

t: $V = 10\ m/s$

$S_A = S_t = S$ -- (1)
$t_A = t_t = t$ -- (2)
$S = \frac{1}{2}(a)t^2$
$S = \frac{5}{2}t^2$
$S = 2.5t^2$ -- (3)

$S = vt$
$S = 10t$ -- (4)

Substitute (3) & (4) on (1)

$2.5t^2 = 10t$
$2.5t^2 - 10t = 0$
$t = \frac{10}{2.5} = \frac{100}{25} = 4\ sec$
$S = 2.5t^2 = 2.5(4)^2 = 2.5(16) = 40\ m$

b. We have $a = 5\ m/s^2$, $t = 4\ sec$, $u = 0$
$V = u + at$
$= 0 + 5(4)$
$= 20\ m/s$

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Use: $g = 10\ m/s^2$

I. ***Choose the best answer from the given alternative.***

A 1. For the object moving in straight line,

A. If the velocity increases by equal amount in equal time interval, the velocity time graph is straight line with positive slope and the acceleration-time graph is horizontal.

B. The slope of displacement time graph and the area of the region under acceleration time graph are, respectively, the acceleration and velocity of the object. ✗

C. If displacement changes by equal amount in equal time interval, the displacement time graph is horizontal line. ✓

D. The area of the region under the displacement time graph and the slope of velocity time graph are, respectively, the acceleration and displacement of the object. ✓

C 2. For an object traveling at a constant velocity

A. average velocity is greater than instantaneous velocity

B. instantaneous velocity is greater than average velocity

C. instantaneous velocity and average velocity are the same

D. instantaneous velocity changes, but average velocity stays the same

3. Under what condition is average velocity equal to the average of the object's initial and final velocity?

$V_{ave} = \frac{u+v}{2}$

A. This can only occur if there is no acceleration.

B. The acceleration is constant.

C. The acceleration must be constantly increasing.

D. The acceleration must be constantly decreasing

A 4. Which of the following statements are true about uniformly accelerated motion?

A. If an object's acceleration is constant, then it must move in a straight line.

B. If an object's acceleration is zero, then it's speed must remain constant.

C. If an object's speed remains constant, then its acceleration must be zero.

D. If the object's direction of motion is changing, then its acceleration is zero ✓

B 5. What does the slope of a velocity-time graph represent?

A. Distance B. Acceleration C. Displacement D. Speed

B 6. A racing car accelerates uniformly from rest along a straight track. This track has markers spaced at equal distances along it from the start, as shown in the figure. The car reaches a speed of 140 km/h as it passes marker 2.



Where on the track was the car when it was traveling at half this speed, that is at 70 km/h?

A. Before marker 1 C. Between marker 1 and marker 2

B. At marker 1 D. After marker 2

C. Two baseballs are thrown from the roof of a house with the same initial speed, one is thrown up, and the other is down. Compare the speeds of the baseballs just before they hit the ground.



A. The one thrown up moves faster because the initial velocity is up

B. The one thrown down moves faster because the initial velocity is down

C. They both move with the same speed

D. The one thrown up moves faster because it has greater acceleration

B 8. The driver of a train travelling at 40m/s applies the brakes as the train enters a station. The train slow down at a rate of 2 m/s. the displacement between initial position and station is

A. 200m B. 400m C. 600m D. 800m

C 9. In a 100m race a person taking 12 seconds to complete the race but a person takes 3 second to attain maximum speed of a person. What is maximum speed?

A. 3.5 m/s B. 5.5 m/s C. 7.5 m/s D. 9.5 m/s

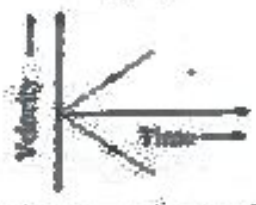
D 10. An object starts from rest at the origin and moves along the x-axis with a constant acceleration of $3m/s^2$. Its final velocity when pass x = 8 m is:

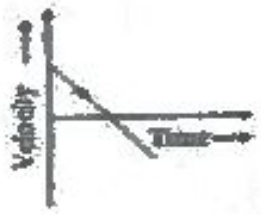
A. 2.5 m/s B. 3.6 m/s C. 5.8 m/s D. 6.9 m/s

D 11. A ball is thrown vertically upward which of the following graph represent velocity time graph of the ball during its flight (neglect air resistance)

A. 

B. 

C. 

D. 

C 12. A ball is thrown down ward from the top of a high cliff. After being released, its down ward

A. velocity will be uniform

B. Acceleration will be greater than g

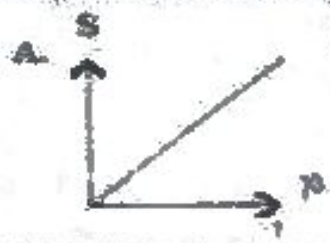
C. Acceleration will be equal to g

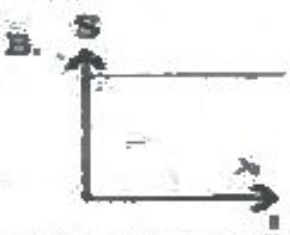
D. Acceleration will be less than g

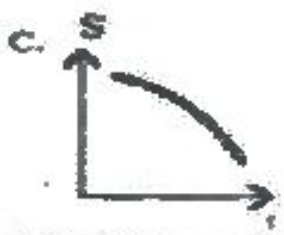
C 13. The velocity of object is $v = (4t^3 + 2) m/s$ and t is in second. Calculate the instantaneous acceleration at 2 second?

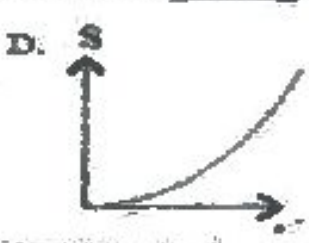
A. $34m/s^2$ B. $45m/s^2$ C. $50m/s^2$ D. $55 m/s^2$

C 14. Which of the following graph is represent uniformly acceleration but in opposite direction?

A. 

B. 

C. 

D. 

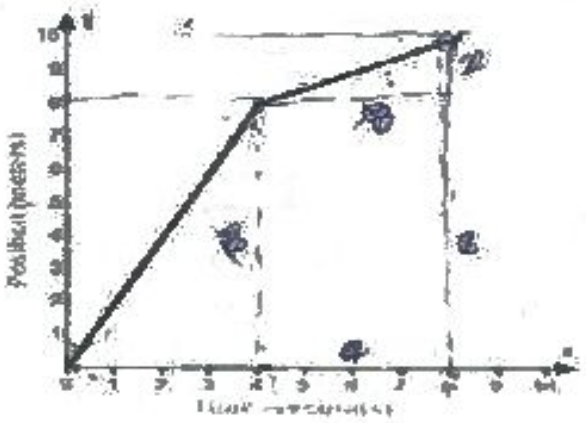
B 15. The position vs time graph of a moving object is shown to the right. What is the average speed from 4 s to 8s?

A. 1 m/s

B. 2 m/s

C. 3 m/s

D. 4 m/s



16. Depend on the above question #15 which of the following velocity vs time graph is equivalent to the above position vs time graph?



A. B. C. D.

17. On its way along a straight line, a car starts from rests from rest and accelerated at a rate of 2.5 m/s^2 for 10 seconds. It maintained its speed for further 5 seconds and decelerated by 0.5 m/s^2 to come to rest. What is the time taken for the entire motion?

A. 65 sec B. 75 sec C. 85 sec D. 90 sec

18. If a car is going 10 m/s and 4 seconds later it is going 18 m/s, what is its acceleration?

A. 2 m/s^2 B. 18 m/s^2 C. 8 m/s^2 D. 2 m/s^2 E. 0.5 m/s^2

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First Semester Physics Test 1 for Grade 11

Dec. 2018 E.C Time Allowed 1:00hr

ANSWER SHEET

Name: Aldonas Mendo Grade: 11 Section: C R.No 33

1. A	8. B	14. C
2. C	9. D	15. B
3. B	10. D	16. A
4. A	11. D	17. A
5. B	12. C	18. A
6. B	13. C	
7. C		